

Falsifiability and Testable Predictions: the $a_0 = c^2 \sqrt{\Lambda/32\pi}$ framework among the major theories of physics

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We place the emergent-gravity framework $a_0 = c^2 \sqrt{\Lambda/32\pi} = 9.36 \times 10^{-11} \text{ m s}^{-2}$ on the same axes as the other leading theories—quantum field theory (the Standard Model), general relativity, Λ CDM, cosmic inflation, standard MOND, Verlinde’s emergent gravity, supersymmetry, string theory, and loop quantum gravity—and ask two operational questions: how many concrete, dated, *falsifiable* predictions does each make, and how many are *confirmed*? The honest answer, stated both ways: quantum field theory and general relativity are the gold standard (richly predictive and confirmed to extraordinary precision); Λ CDM is the confirmed concordance model carrying real tensions; the present framework and standard MOND are highly falsifiable but only partially confirmed, sitting in the “testable-not-yet-confirmed” regime with conceded losses; and string theory makes essentially zero unique near-term confirmable predictions, drawing the standard falsifiability critique. The framework is decisively *more falsifiable* than string theory—but that is a low bar, it is not more predictive than the Standard Model, and its one genuinely distinctive prediction (a declining $a_0(z)$) is currently unconfirmed and leaning unfavourable. We give the comparative scorecard, the falsifiability–confirmation map, and the per-theory prediction lists.

I. WHAT “TESTABLE” MEANS HERE

A theory earns scientific standing by making predictions that could have come out otherwise and then survived the test. We score each theory on three operational axes: (i) the number of *concrete, falsifiable* predictions it makes—predictions with a stated threshold that a measurement could violate; (ii) how many of those are *confirmed*; and (iii) its *falsifiability*—whether a feasible experiment could, even in principle, kill it. These axes are qualitative characterisations, not exact data, and we flag them as such; but the gross ordering they produce is not in serious dispute. Crucially, “more falsifiable than theory X ” is *not* the same as “more correct than X ,” and we keep the two separate throughout.

II. THE COMPARATIVE SCORECARD

Table I summarises. The gradations are deliberately coarse.

III. THE FALSIFIABILITY–CONFIRMATION MAP

Figure 1 plots each theory by how falsifiable it is (horizontal) against how confirmed (vertical). The upper right is occupied by the Standard Model and general relativity—sharp *and* verified. The present framework and standard MOND sit lower right: highly falsifiable, partially confirmed. String theory sits lower left, near-unfalsifiable and unconfirmed. The framework’s position is the honest one for a young, testable, not-yet-established theory—clearly to the right of string theory, clearly below the confirmed giants.

TABLE I. Testable predictions across the major theories. “Predictions” counts concrete falsifiable claims; “confirmed” is the track record to date; falsifiability is graded $\star$ (effectively unfalsifiable near-term) to $\star\star\star\star$ (sharp, survived every test).

Theory	Pred.	Confirmed	Falsifiability
QFT / Standard Model	~ 25	nearly all	$\star\star\star\star\star$
General relativity	~ 10	all to date	$\star\star\star\star\star$
Λ CDM	~ 9	most	$\star\star\star\star$
Cosmic inflation	~ 4	spectrum only	$\star\star\star$
This framework	$\sim 9^\dagger$	galaxy RAR (shared)	$\star\star\star\star$
Standard MOND	~ 8	galaxy-scale yes	$\star\star\star\star$
Verlinde emergent grav.	~ 3	lensing mixed	$\star\star\star$
Supersymmetry	~ 5	none (absent)	$\star\star\star$
Loop quantum gravity	~ 2	none	$\star\star$
String theory	~ 0 unique	none	$\star$

$^\dagger \sim 9$ dated falsifiable, of which only ~ 3 are *distinctive* (beyond standard MOND); all distinctiveness flows from the single $a_0(z) \propto \sqrt{\rho_{DE}}$ spine.

IV. PER-THEORY PREDICTIONS

QFT / Standard Model (the ceiling). The electron anomalous moment is verified to ~ 0.13 parts per trillion—the most precise prediction in science; plus the Higgs (predicted, found 2012), the W/Z bosons, neutrino oscillations, running couplings, CP violation, the Lamb shift, the Casimir effect, and a host of masses and cross-sections. Nothing else listed approaches this; the framework does not claim to.

General relativity. Perihelion precession, light bending, gravitational redshift, the Shapiro delay, frame-dragging, binary-pulsar decay, gravitational waves (2015), the black-hole shadow (2019)—all confirmed.

Λ CDM. The CMB acoustic-peak spectrum, big-bang

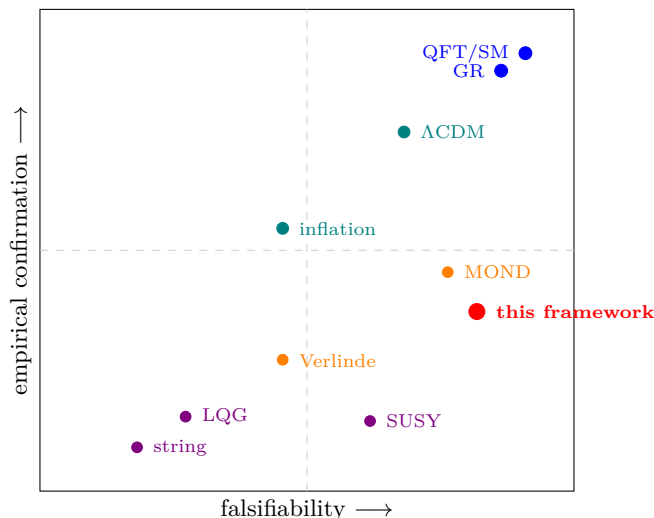


FIG. 1. Each theory placed by falsifiability (right = more) and empirical confirmation (up = more). Quantum field theory and general relativity occupy the tested-and-confirmed corner; the present framework and MOND are highly falsifiable but only partially confirmed; string theory is near-unfalsifiable and unconfirmed. Axes are qualitative.

nucleosynthesis, the baryon acoustic scale, structure growth, lensing maps—broadly confirmed, but the dark-matter particle is undetected after forty years and Λ 's value is unexplained, and it carries the $\sim 5.8\sigma$ Hubble tension and a $\sim 3\sigma$ preference for evolving dark energy.

This framework. ~ 9 dated, falsifiable predictions: the declining $a_0(z) \propto \sqrt{\rho_{DE}}$ (DESI DR3, high- z kinematics); the high- z baryonic Tully–Fisher offset; the wide-binary external-field boost (+6.6% in velocity on the framework's own de Sitter–Unruh law, Gaia DR4); the external field effect itself; the lensing radial-acceleration relation; a Solar-System (Cassini) bound; a predicted a_0 bump near $z \approx 0.4$. The radial-acceleration relation and BTFR successes are *shared with standard MOND*; the distinctive content is the $a_0(z)$ evolution—currently leaning $\sim 1.5\sigma$ unfavourable.

Standard MOND. The radial-acceleration relation (a real, tight success), the BTFR, the external field effect, rotation-curve shapes—confirmed at galaxy scale; fails galaxy clusters ($\sim 2\times$) and the CMB.

Supersymmetry. Superpartners, gauge-coupling unification, a light Higgs. Superpartners are *not* found at the LHC (gluinos excluded to ~ 2.2 TeV); the one pre-

diction that landed (a light Higgs) is shared with the Standard Model.

String theory. Requires supersymmetry (absent) and extra dimensions (absent); “predicts gravity” is a post-diction; the $\sim 10^{500}$ -vacua landscape yields no unique near-term confirmable test, which is the substance of the long-standing falsifiability critique.

Loop quantum gravity. Discrete spacetime, a possible Planck-scale Lorentz violation (searched in gamma-ray-burst timing, not seen), a big-bounce cosmology—no near-term confirmation.

V. THE HONEST VERDICT

Versus string theory. The framework is decisively more falsifiable: ~ 5 – 6 dated, telescope-named, kill-thresholded near-term predictions against essentially zero unique confirmable string predictions. This is a genuine virtue—risky, dated bets where the alternative makes none. *Three guardrails, each true and each stated:* (i) falsifiability is precisely the axis the framework wins on; (ii) that is a *low bar*—beating string theory on testability is not being established; (iii) more predictions $\neq$ more correct, and the framework is *not* more predictive than the Standard Model, whose electron $g-2$ stands at 0.13 ppt.

Where the framework actually stands. It is a falsifiable, internally consistent, Λ -anchored modified-inertia realisation that reproduces MOND's $z=0$ galaxy record, is sign-aligned with DESI's evolving dark energy (the one cosmological tension that points its way), and makes a sharp distinctive bet in $a_0(z)$. It also carries real, conceded losses—chief among them an a_0 -independent 8.6 – 9.2σ weak-lensing morphology split that no choice of convention removes—and an underived coefficient $Z = \sqrt{32\pi/3}$. It does not break Λ CDM, and Λ CDM does not falsify it; the verdict comes from the 2026–2032 forward tests.

Bottom line. On the one axis that most cleanly separates science from speculation—can a measurement kill it?—the framework is among the strongest of the live alternatives and far ahead of string theory. On the axis that separates a hypothesis from established physics—is it confirmed?—it sits, honestly, with MOND in the testable-but-unproven quadrant, well below quantum field theory and general relativity. Both statements are true at once, and keeping them distinct is the whole point.

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